

PHYSICS PRACTICAL SHEETS

Date: 20/12/11

PN CAMPUS

Class: BSc 2nd year

Roll No.: 68

Experiment No.: 13

Shift: Day

Group: P₂

Sub: Physics

Object of the Experiment (Block Letter)

Set: 1

OBJECT: TO DETERMINE THE SPECIFIC ROTATION OF SUGAR SOLUTION USING LAURENT'S HALF SHADE POLARIMETER

APPARATUS REQUIRED:

- | | |
|--------------------------------------|-------------------|
| (i) Laurent's half-shade polarimeter | (iii) Sugar |
| (ii) Sodium lamp | (iv) A weight box |
| (iii) A balance | (v) Beaker |
| (iv) Graduated cylinder | (vi) Filter paper |
| (v) Funnel | |

THEORY:

If ' θ ' is the optical rotation produced by ' l ' decimeters of a solution and ' c ' the concentration in gram per cubic centimeter, then specific rotation's at given temperature ' t ' and corresponding to a wavelength λ is given by,

$$[\alpha]_{\lambda}^t = \frac{\theta}{lc} = \frac{\text{Rotation in degree}}{\text{length in decimeter} \times \text{concentration in gm/cc}}$$

$$s = \frac{\theta}{lc}$$

OBSERVATION:

Weight of sugar (m) = 20 gm

Value of 10 division of main scale = 10°

Value of 1 division of main scale = 1°

10 division of V.S. coincides with 9 divisions of M.S.

1 division of V.S. coincides with $\frac{9}{10}$ divisions of M.S.

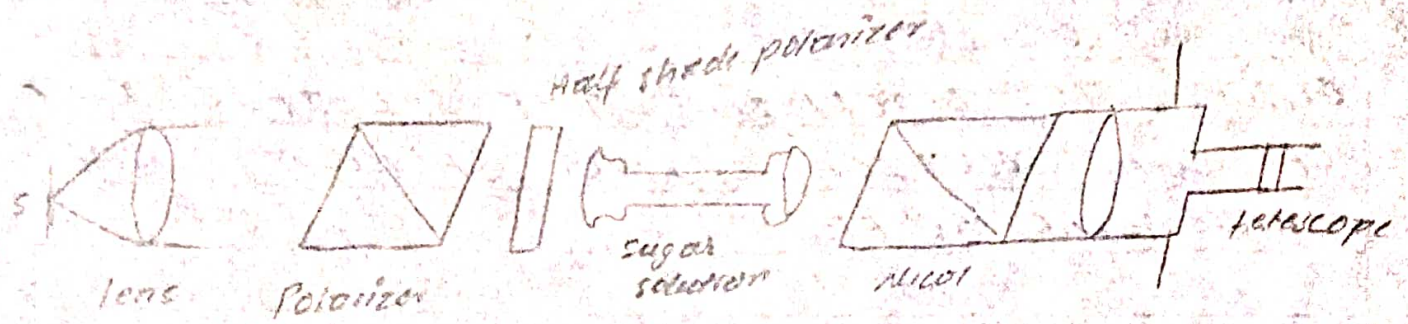
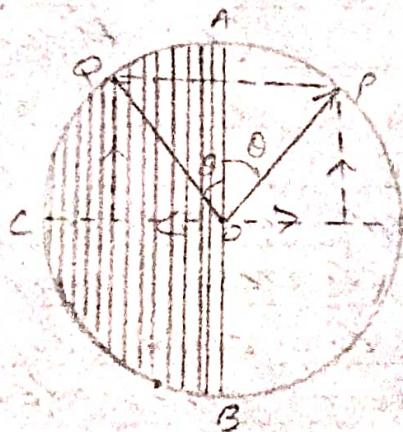


Fig: Laurent's half shade polarimeter



$$\therefore \text{Value of 1 V.S.d} = \frac{9}{10} \times 1 \text{ M.S.d}$$

$$= \frac{9}{10} \times 1^\circ = \frac{9}{10}^\circ$$

$$\text{Verrier constant (V.C.)} = 1 \text{ M.S.d} - 1 \text{ V.S.d} \\ = 1^\circ - \frac{9}{10}^\circ$$

$$\text{V.C.} = \frac{1}{10}^\circ$$

OBSERVATION TABLE:

S.N	Strength of solution		Scale reading through sol ⁿ :						Rotation		Mean θ
	in gm per cc.		1 st position			2 nd position			1 st	2 nd	
	100 cc	10 cc	M.S.	V.S.	Total	M.S.	V.S.	Total	position	position	
1.	0	0	43	3	4.3	184	1	184.1	θ_1	θ_2	
2.	20	0.2	26	0	26	205	5	205.5	21.7	21.4	21.55
3.	10	0.1	14	7	14.7	193	5	193.5	10.4	9.4	9.9
4.	5	0.05	9	9	9.9	191	4	191.4	5.6	7.3	6.45

$$l \Rightarrow 10 \text{ cm} = 1 \text{ dm}$$

$$\Rightarrow 20 \text{ cm} = 2 \text{ dm}$$

$$\text{Specific rotation } (S_1) = \frac{\theta}{lc} = \frac{21.55}{2 \times 0.2} = 53.875 \text{ degree dm}^{-1} \text{ gm}^{-1} \text{ cc}$$

$$\text{Specific rotation } (S_2) = \frac{\theta}{lc} = \frac{9.9}{2 \times 0.1} = 49.5 \text{ degree dm}^{-1} \text{ gm}^{-1} \text{ cc}$$

$$\text{Specific rotation } (S_3) = \frac{\theta}{lc} = \frac{6.45}{2 \times 0.05} = 64.5 \text{ degree dm}^{-1} \text{ gm}^{-1} \text{ cc}$$

$$\therefore \text{Mean specific rotation } (S) = \frac{S_1 + S_2 + S_3}{3}$$

$$= \frac{53.875 + 49.5 + 64.5}{3}$$

$$= 55.958^\circ \text{ dm}^{-1} \text{ gm}^{-1} \text{ cc}$$

RESULT:

Hence, the specific rotation of sugar was found to be $55.958^\circ \text{ dm}^{-1} \text{ gm}^{-1} \text{ cc}$.

PRECAUTIONS AND SOURCE OF ERROR:

- (i) The polarimeter tube should be well cleaned.
- (ii) Water and sugar used should be dust free.
- (iii) Whenever a solution is changed, rinse the tube with new solution under construction or clean the tube with water and dry it.
- (iv) There should be no bubble inside the tube.
- (v) The position of analyzer should be set accurately.
- (vi) Readings should be taken when halves of the field of view become equally illuminated.